

Simultaneous equations



UNIT 4: The method of substitution

QUESTION 1 Solve the following pairs of simultaneous equations by substitution.

a $x + y = 10$
 $y = x - 8$

b $2p - q = 12$
 $p = 3 - q$

c $x + 3y = 15$
 $y = x + 1$

d $x + 4y = 21$
 $x = 12 - y$

e $y = 6 - x$
 $2x - y = -6$

f $2x + y = 7$
 $x = 4 - y$

QUESTION 2 Use the method of substitution to solve the following pairs of simultaneous equations.

a $2m + 3n = 8$
 $3m + 3n = 5$

b $2x + 3y = 12$
 $4x - 3y = 6$

c $2x - 5y = 11$
 $2x - 3y = 7$

d $y = 2x + 1$
 $y = x + 4$
